Reflection of the paper:
LLM Serving Optimization with Variable Prefill and Decode Lengths

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Abstract

This is the reflection of the paper after a deeper review that was mentioned during the previous meeting. Link of the Paper is in here.

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1 Reflections and Insights

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1.1 General

The optimization techniques proposed for LLM serving hold substantial value for industrial applications from my perspective, especially combinatorial optimization. While the current market heavily emphasizes model accuracy and the size of the trained models, optimizing operational latency (TEL) is critical for improving user experience and minimizing hardware expenditures, such as memory overhead. This paper provides specific insights into balancing latency reduction with algorithmic complexity, offering a highly practical framework for real-world deployment.

1.2 Elegance Proof of the Upper Bound of Competitive Ratio

The theoretical approach used to prove the upper bound of the competitive ratio is exceptionally creative and rigorous. Rather than attempting a direct, brute-force construction of the worst-case instance to maximize the ratio $\frac{\text{TEL(Sorted-F;}I)}{\text{TEL(Optimal;}I)}$, the paper cleverly introduces a multi-tiered strategy. By constructing intermediate algorithmic variants from Sorted-F (*Separate*, *Group*, and *Align*) and also constructing $\text{Optimal}_{\text{Align}}$, the proof establishes an elegant chain of inequalities that smoothly bounds the competitive ratio with the final inequality TEL(Sorted-F;I)$<$ 48TEL(Optimal; I). Sustaining such a lengthy

and intricate bounding chain demonstrates clear and remarkable mathematical intuition and analytical ingenuity, particularly in designing specific variations that prevent the theoretical upper bounds from diverging.

1.3 Robustness of the Algorithm

The empirical evaluations demonstrate that the proposed algorithms are highly robust and applicable to a wide range of the amounts of requests in realistic workloads. Notably, the heuristic variants of Sorted-F (Sorted by Local Swap and Quantile Greedy Selection) and the Relaxed Linear Programming yield similar performance and robustness on real world data and are competitive on statistically synthetic distributions, even the time complexity of variants ALGs from Sorted-F is much less than solving the integer programming in the Relaxed-LP. Furthermore, the paper provides clear guidelines on selecting the optimal algorithmic variant based on the request volume (n), providing clear instructions for LLM developers.

2 Problems unresolved

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2.1 Overview

While I believe I have grasped the foundational proofs and structural arguments established in the main text, a few intricate mathematical transitions in the Appendix remain challenging to fully reconcile.

2.2 Proof of Thm. EC. 1

In ec1 to ec2 of the paper (P. 35∼36) (the link direct to the exact location of the proof in the paper). Thm. EC. 1 is as followed.

Theorem EC.1. Suppose that each request i ∈ [n] satisfies $s_i \in$ [1,M], $o_i \in$ [1,M] and $s_i + o_i \leq M$, then Algorithm MC-SF2 (process the smallest $s_i + o_i$ first) achieves an unbounded competitive ratio. Specifically, as $M \to \infty$, we have CR(MC-SF$_2$) $\to \infty$.

This theorem show that besides only when sorted by o_i in Thm. 1, the competitive ratio of sorting by $s_i + o_i$ is already unbounded even when $\frac{\text{TEL(MC-SF}_2\text{; I)}}{\text{TEL}(\mathcal{A}\text{; I})}$, even the TEL of the denominator is not the smallest by $\mathcal{A}$ could be not the Optimal Algorithm.

Though I have obtained the same formula for TEL(MC-SF$_2$; I) and TEL($\mathcal{A}$; I) from the paper by assuming $\sqrt{M} - 1 = \sqrt{M}$ or $\sqrt{M} + 1 = \sqrt{M}$ when M is large. The steps in the paper are remains unclear for me. I have difficulties to get the exact term of

$$\frac{\text{TEL(MC-SF}_2\text{; I)}}{\text{TEL}(\mathcal{A}\text{; I)}} = \frac{\text{TEL(MC-SF}_2\text{; } I)}{\text{TEL}(\mathcal{A}\text{; } I)} = \frac{\frac{1}{2}M^2 + M^{2.5} + \frac{1}{2}M^{1.75}}{\frac{1}{2}M^2 + M^2 + \frac{1}{2}M^{1.75}}$$

Instead, here is my steps after getting the TEL for both MC-SF$_2$ and $\mathcal{A}$ with same setting of data X and Y, $|X| = M$ and $|Y| = M^{1.5}$.

$$\begin{aligned}
\text{TEL(MC-SF}_2\text{; I)} &= (\sqrt{M}-1)\sqrt{M}\sum_{i=1}^{\frac{X}{\sqrt{M}}} i + (\sqrt{M}-1)\frac{XY}{\sqrt{M}} + \sqrt{M}\sum_{j=1}^{\frac{Y}{\sqrt{M}}} j \\
&= (\sqrt{M}-1)\sqrt{M}\frac{(\sqrt{M}+1)\sqrt{M}}{2} + (\sqrt{M}-1)M^2 + \sqrt{M}\frac{(M+1)M}{2} \\
&= \frac{M^2 - M}{2} + M^{2.5} - M^2 + \frac{M^{2.5} + M^{1.5}}{2} \\
&= \frac{3M^{2.5} - M^2 + M^{1.5} - M}{2}
\end{aligned}$$

$$
\begin{aligned}
\mathrm{TEL}(\mathcal{A};\mathrm{I}) &= (\sqrt{M}-1)\sqrt{M}\sum_{i=1}^{\frac{X}{\sqrt{M}}} i + \frac{XY}{\sqrt{M}} + \sqrt{M}\sum_{j=1}^{\frac{Y}{\sqrt{M}}} j \\
&= (\sqrt{M}-1)\sqrt{M}\frac{(\sqrt{M}+1)\sqrt{M}}{2} + M^2 + \sqrt{M}\frac{(M+1)M}{2} \\
&= \frac{M^2-M}{2} + M^2 + \frac{M^{2.5}+M^{1.5}}{2} \\
&= \frac{M^{2.5}+3M^2+M^{1.5}-M}{2} \\
\implies \mathrm{CR}(\text{MC-SF}_2) \geq \frac{\mathrm{TEL}(\text{MC-SF}_2;\mathrm{I})}{\mathrm{TEL}(\mathcal{A};\mathrm{I})} &= \frac{3M^{2.5}-M^2+M^{1.5}-M}{M^{2.5}+3M^2+M^{1.5}-M} \\
&= \frac{3-\frac{1}{M^{0.5}}+\frac{1}{M}-\frac{1}{M^{1.5}}}{1+\frac{3}{M^{0.5}}+\frac{1}{M}-\frac{1}{M^{1.5}}} \\
&= 3 \qquad \text{when } M \to \infty
\end{aligned}
$$

, which I am not sure if I missed an implicit scaling relationship or a specific constraint between the dimensions X, Y, and M that allows the exponent to decouple and dominate the denominator as presented in the paper?

2.3 Proof of Lemma 3

In ec5 of the paper (P. 39), I am wondering about if the second last line of the proof is really the TEL(Sorted-F$_{seperate}$; I).

Here is the original Lemma 3.

$$
\mathrm{TEL}(\text{Sorted-F}_{\text{group}}) \geq \frac{1}{4}\mathrm{TEL}(\text{Sorted-F}_{\text{seperate}})
$$

Instead, after line 4 of the proof, which is

$$
\mathrm{TEL}(\text{Sorted-F}_{\text{group}}) > \frac{1}{4}\sum_{m=1}^{\infty}\Big(\sum_{l=b_{m-1}+1}^{b_m} o_{l,n_l}\big(\sum_{k=b_m+1}^{\infty} n_k + \sum_{k=b_{m-1}+1}^{b_m} n_k\big)\Big) + \sum_{k=1}^{\infty}\sum_{i=1}^{n_k} o_{k,i}
$$

here is my approach after line 4,

$$> \frac{1}{4}\sum_{m=1}^{\infty}\Big(\sum_{l=b_{m-1}+1}^{b_m} o_{l,n_l}\sum_{k=b_m+1}^{\infty} n_k + \sum_{l=b_{m-1}+1}^{b_m-1} o_{l,n_l}\sum_{k=l+1}^{b_m} n_k\Big)+\sum_{k=1}^{\infty}\sum_{i=1}^{n_k} o_{k,i}$$

$$> \frac{1}{4}\Big(\sum_{m=1}^{\infty}\Big(\sum_{l=b_{m-1}+1}^{b_m} o_{l,n_l}\sum_{k=b_m+1}^{\infty} n_k + \sum_{l=b_{m-1}+1}^{b_m-1} o_{l,n_l}\sum_{k=l+1}^{b_m} n_k\Big)+\sum_{k=1}^{\infty}\sum_{i=1}^{n_k} o_{k,i}\Big)$$

$$> \frac{1}{4}\text{TEL}(\text{Sorted-F}_{\text{seperate}})$$

However, based on the standard definition of TEL(Sorted-$\text{F}_{\text{separate}}$), the precise intra-group cross-batch cascading delay requires that each batch X_l imposes a specific waiting cost on subsequent intra-group batches $X_{l+1}, \ldots, X_{b_m}$:

$$\text{TEL}(\text{Sorted-F}_{\text{seperate}}) = \sum_{m=1}^{\infty}\Bigg(\sum_{l=b_{m-1}+1}^{b_m} o_{l,n_l}\sum_{k=b_m+1}^{\infty} n_k + \sum_{l=b_{m-1}+1}^{b_m-1}\Big(o_{l,n_l}+(b_m-l)\Big)\Bigg)+\sum_{k=1}^{\infty}\sum_{i=1}^{n_k} o_{k,i}$$

seems not showing the latency increased to each request in batch X_{l+1} brought by the batch X_l, l$\in [b_{m-1}+1, b_m-1]$ in the term of

$$\sum_{l=b_{m-1}+1}^{b_m-1}\Big(o_{l,n_l}+(b_m-l)\Big)$$

in the final line.

3 Resolved Queries and Inspirations

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3.1 Overview

This section outlines the conceptually enriching queries that arose during my deep review. While navigating the intricate formulations presented several structural challenges, these have been successfully resolved through careful deduction. To maintain conciseness, this section focuses strictly on the insights and extensions that are more inspirational and

tactical derived from the text.

3.2 Resolved Queries

Q1: Rather than swapping a single request into or out of the candidate batch sequentially, could the Local Swap search be extended to evaluate higher-order combinations (e.g., swapping a pair of tightly coupled requests in the batch simultaneously with a single larger request that is not in the batch)?

A1: Yes, but the complexity of ALG would be much higher by adding one for loop for combinations of choosing 2 requests that are in the batch. Also aware that in F-metric we would also prefer more requests processing, then it is relatively less common to swap the single request not in the batch with more than 1 requests in the batch at the same time.

Q2: In P. 25 Table 3, there are recommended scale for different Algorithm including Exact Programming ($n \leq 100$), Local Swap Search ($n \leq 500$), Quantile Greedy Selection ($n \geq 1000$), which algorithm to use when $500 < n < 1000$?

A2: Here are 2 recommendations.

1. Requests sorted by $s_i + o_i$. Capture only first 500 sorted requests and use Local Swap Search and leave other to the other X afterwards.

2. If $500 < n < 750$, use Local Swap Search. If $750 \leq n < 1000$, use Quantile Greedy Selection.

3.3 Consolidated Knowledge and Theoretical Takeaways

1. Meaning of Heuristic ALG: A relaxed ALG that the solution is less optimized.

2. Complexity of solving Linear Programming is related to LU decomposition in Gurobi, which is learned from my coursework in Numerical Method course.

3. Use Linear Programming to look for the expected value of the optimization.

4. Knapsack Problem.

5. 3-Partition problem is strong NP-complete problem.

6. Combinatorial optimization have many NP-hard problem when there are a lot of combinations of the solutions.

7. Use square of $|X|$ as the denominator in F-metric to also optimize the amounts of requests.

8. Rasied awareness that the prefill tokens in the same request could be processed at the same time and decode tokens in the same request need to be generated one-by-one by Auto-regressive.

9. Resource-Constrained Project Scheduling problem.

10. 2D Bin Packing Problem.

11. I recognized the critical importance of Stratified Sampling during experiments. For example, in P. 30, When subsampling from the pooled trace (composed of 1600 short conversations and 400 long summarizations), keeping a strictly proportional task distribution across all test sizes (200 to 2000) is mandatory to neutralize experimental bias and prevent statistical drift.

12. Competitive Ratio, offline and online Algorithm, though it is first learn from the other paper written by the other Professor.

3.4 Other Inspirations

1. Could the optimization be also studied in scheduling the requests in image generations, or mixed mode with generating texts and images? For some image generation software, they are not using auto-regressive mechanism by generating the token one by one, they

are using the diffusion model, could it also be studied?

2. Could the paper combined with the probability of amount of requests arrive to the system at certain time and become an inventory control problem in Stochastic Modeling? It is inspired by this paper written by the other Professor.

4 Conclusion and Acknowledgments

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In summary, this paper makes a landmark contribution to the field of LLM serving optimization. While contemporary research predominantly focuses on accelerating single-model accuracy or structural optimizations within specific neural network layers, this work instead introduces a macroscopic perspective to optimize the LLM processing time. By formulating the request scheduling problem from an operations research perspective, it elegantly minimizes system-wide latency and operational overhead.

The comprehensive analysis detailed throughout this paper has significantly enriched my theoretical knowledge and reshaped my understanding of hardware-software co-design. I would like to express my sincere gratitude to you for providing this profound research framework and for taking the time to review this reflection.

5 References

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